



Figure 1: Simulation summary



Figure 2: Data centre characteristics

Multiple simulations can be performed using a single run script. In that case, the results are plotted as a tabbed pane.

### To modify the existing source code

The above examples show the parameter changes in the existing network and how to analyse the results. However, if a researcher is trying to configure a CPU or a HPC cluster, alter cache memory, handle virtual machines, etc, then there should be code modification in the source files (.cc and .h).

These files are located in the `~greencloud/build/ns-2.35/greencloud/` directory and are already compiled as object files. Any changes to these files need a compilation as specified below:

Once the `cpu.cc` file is modified, it will be compiled using the make command as shown below.

```
-pradeep@localhost $] cd /home/pradeep/greencloud/build/ns-2.35/
-pradeep@localhost ns-2.35 $] make
```

If new set of files (newfile.cc and newfile1.cc) are added, those details have to be added to `ns-2.35/Makefile.in` as specified below in the `OBJ_CC` variable. For each .cc file, there need to be a .o file to be added.

```
OBJ_CC = \
....
greencloud/newfile.o \
greencloud/newfile1.o \
```

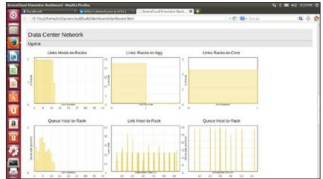


Figure 3: Data centre network characteristics



Figure 4: Energy consumption details

Table 1: Simulation data

| Data centre architecture | Three tier debugging |
|--------------------------|----------------------|
| Core switches            | 1                    |
| Aggregation switches     | 2                    |
| Access switches          | 2                    |
| Number of servers        | 40                   |
| Users                    | 1                    |
| Average load/server      | 0.3                  |
| Total tasks              | 688                  |
| Average task/server      | 17.2                 |
| Total energy calculated  | 322.7 watt hour      |
| Server energy            | 164.1 watt hour      |
| Total switch energy      | 158.6 watt hour      |

GreenCloud is the best open source tool to analyse the performance of a data centre. The parameters of the cloud can be varied, and it comes with a provision to add or modify existing source code to define new metrics for a cloud. Any questions on installation or tuning GreenCloud are always welcome. 

### References

[1] <http://greencloud.gforge.unl.nl/>

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